

1.

What is the purpose of the new operator in C++?

- A. To allocate memory on the stack
- B. To allocate memory on the heap
- C. To deallocate memory
- D. To initialize variables

Answer: B

2.

What is the problem with the following code snippet?

```
#include<iostream>
using namespace std;
int main(void)
{
    int* ptr = new int[10];
    delete ptr;
    ptr=NULL;
    cout<<"Memory is freed"<<endl;
    return 0;
}
```

- A. it will work correctly
- B. Dangling pointer
- C. Double deletion
- D. Incorrect deallocation for dynamic array

Answer: D

3.

How can we fix the issue in the following code snippet to perform a deep copy?

```
#include<iostream>
using namespace std;
class Point
{
    private:
        int* ptr;
    public:
        Point (int value)
        {
            ptr = new int;
            *ptr = value;
        }
        Point (const Point & obj)
        {
            ptr = new int;
            *ptr = *obj.ptr;
        }
};
```

- A. ptr = obj.ptr;
- B. ptr = new int; *ptr = *obj.ptr;
- C. ptr = new int; ptr = *obj.ptr;
- D. *ptr = obj.ptr;

Answer: B

4.

When should you implement a copy constructor?

- A. When the class has no dynamically allocated memory**
- B. When the class has dynamically allocated memory**
- C. When the class has static data members**
- D. When the class has const data members**

Answer: B

5.

What is the benefit of performing a deep copy instead of a shallow copy?

- A. It reduces memory usage**
- B. It improves performance**
- C. It prevents double deletion and ensures data integrity**
- D. It simplifies the code**

Answer: C